FATİH ÖZTÜRK

Mechatronics Engineer | Robotics, Embedded Systems & Computer Vision

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PROFESSIONAL SUMMARY

Mechatronics engineer who takes systems from architecture to working hardware - mechanical design, custom electronics, real-time control and vision algorithms. Designed, built and validated a TÜBİTAK-funded robotic test platform, and led a 20-person research team.

EDUCATION

Yıldız Technical University - B.Sc. Mechatronics Engineering (100% English) Graduated July 2026

GPA 3.47 / 4.00 · **Ranked 3rd in the graduating departmen of Mechatronics Engineering** · Graduated with Honours

SELECTED PROJECTS

ARC-HIT - Autonomous Robotic Archery Test Platform · Demonstration video 09/2025 - 06/2026

Graduation Project · Team Leader · Supported by TÜBİTAK 2209-B

- Designed the overall system architecture of a robotic archery platform that removes the human variable from arrow ballistic testing, making shot-to-shot comparison repeatable.
- Electronic system design: a two-layer architecture separating high-level vision processing from real-time motion control, on a custom PCB with isolated power rails that keep motor noise off the sensor and logic lines.
- Control system design: closed-loop positioning of the aiming axes driven by visual feedback, plus controlled bow-draw and release, modelled and tuned in simulation.
- Algorithm design: an image processing pipeline extracting the target centre and post-shot impact point as an angular error vector, plus automated shot-group statistics.
- Verified against the design requirements: 42 mm average 150-shot group diameter at 18 m, against an ≤80 mm target.

Tools: Embedded C, Python/OpenCV, KiCad, MATLAB/Simulink, SolidWorks

AI-Based Athlete Posture Analysis System 12/2024 - 06/2025

- Built a MediaPipe-based pipeline fusing video from 3 synchronised cameras to extract body joint angles at the moment of release.
- Designed and manufactured the telescopic camera platform in SolidWorks (2 m spacing, up to 2.50 m height).
- Developed a Unity interface mapping the measured angles onto a 3D avatar for shot review.

Tools: Python, OpenCV, MediaPipe, Unity, SolidWorks

TEKNOFEST 2025 - Artificial Intelligence in Healthcare 02/2025 - 08/2025

- Developed deep learning models detecting ischemic and haemorrhagic stroke from non-contrast CT, plus onset-timing prediction from diffusion MRI (Python, PyTorch).

WORK EXPERIENCE

Advanced Systems and Invention Laboratories (ASIL) - Istanbul, Türkiye 09/2023 - 07/2026

Undergraduate Student Team Leader (07/2025 - 07/2026) · Undergraduate Researcher (09/2023 - 06/2025)

- Led 20 undergraduate researchers in a multidisciplinary robotics and AI lab: ran recruitment interviews and designed technical training plans.
- Owned the end-to-end system design of ARC-HIT as project lead, with a teammate on the mechanical workstream.
- Applied machine learning, deep learning , CNN and object detection (YOLO) across lab research projects.

Enlighter Medical Device Technologies - Istanbul, Türkiye 01/2026 - 02/2026

R&D Engineer (Intern)

- Designed and integrated subsystems of a CNC-based PCB prototyping machine for in-house prototyping of medical device boards.

TAAC Aerospace Technologies - Istanbul, Türkiye 07/2025 - 08/2025

Embedded Systems Engineer (Intern)

- Developed bare-metal STM32 firmware at register level, including encrypted data transmission between nodes.
- Inspected hardware in the lab to verify node-to-node communication and validate firmware on real targets.
- Designed a CAN bus data-transfer hub intended for use in the KAAN aircraft, defining the node topology and message routing for the onboard link.

PUBLICATIONS

Classification of Normal and Abnormal Heartbeats via R-Peak-Based Segmentation of ECG Signals - medRxiv preprint, 2025.

CNN-Based Detection of Cardiovascular Diseases from ECG Images - arXiv preprint, 2024. InceptionV3, 93.27% accuracy.

TECHNICAL SKILLS

Programming: Python · C · C++ · MATLAB/Simulink · Linux

Robotics & Embedded: ROS2 · STM32 (HAL) · CAN Bus · KiCad · Proteus · PLC

AI & Vision: Deep Learning · CNN · Object Detection (YOLO) · OpenCV · MediaPipe · Keras · TensorFlow · PyTorch

CAD/CAM: SolidWorks · Fusion 360 · CNC/CAM · 3D printing **Languages:** Turkish (native) · English (advanced)